P3Q Corona Kind: Module-LWE Threshold Verify on the Unified Precompile Slot

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Abstract

LP-218 introduces P3Q at EVM slot `0x012205` as the PQ-verifier family for rollup-batch authentication, specifying the unified kind-byte dispatch ABI and the first verifier kind $0x01$ (Pulsar / FIPS-204 ML-DSA-65 threshold). This paper specifies *kind* $0x02$ *Corona*: a Module-LWE threshold verifier built on the two-round Corona construction (eprint 2024/1113). The shared P3Q framework — the unified slot, the kind-byte dispatch ABI, and the `P3QVerifier` registration interface — is specified in LP-219; this paper gives Corona's wire layout and its gas calibration (5 000 000 gas). Corona reduces to Module-LWE (Ringtail/Raccoon lineage). It shares the Module-LWE *hardness family* with Pulsar (LP-223) but is a distinct scheme at a distinct parameter set: the dual-lattice posture forces an adversary to break *both* constructions — strictly stronger than either alone, though *not* hardness-family-disjoint, since both reduce to Module-LWE. Cross-family independence comes only from Magnetar's hash-collision assumption (LP-222). Corona is the Module-LWE parameter-diversity leg of the LP-217 PQ-strict posture. We also place Corona in the DEX rollup settlement pipeline: per-market fill batches roll up to the Z-Chain and are verified on a two-rung ladder — a kind $0x01$ ML-DSA-65 attestation rung (measured ∼4714 verifies/s) and a `starkfri` STARK/FRI validity rung (LP-221, measured ∼986/s on a toy AIR) — with Corona as the optional Module-LWE second leg for value-bearing markets and bundled $K \times V \times$ markets settlement. GPU dispatch is projected, not measured. Corona's threshold library has landed (`corona v0.10.x`; the dedicated Corona precompile at `0x012206` is wired); the *P3Q-family* kind-$0x02$ dispatch entry at `0x012205` remains reserved and returns `abiFalse` via `ErrKindNotImplemented` until that dispatch is wired.

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1 Introduction

P3Q (LP-219) is the unified-slot post-quantum verifier framework at EVM precompile slot `0x012205`: one slot, one ABI, and a leading `kind` byte that dispatches into a per-kind threshold-signature verifier core. This paper specifies *kind* $0x02$ *Corona*, the Module-LWE member of the P3Q family.

For the slot allocation, the kind-byte dispatch ABI, the `P3QVerifier` Go interface that registers each kind, and the cross-family composition into LP-217 cert profile modes, see LP-219. The sibling kinds are specified in LP-223 (Pulsar / rank-k Module-LWE) and LP-222 (Magnetar / hash-based FIPS-205 SLH-DSA).

Corona reduces to the hardness of Module-LWE (Ringtail/Raccoon lineage, built on the two-round construction of eprint 2024/1113). It shares the Module-LWE *hardness family* with its sibling Pulsar (kind $0x01$) — the two differ in parameter regime and codebase, not hardness family, so pairing them buys construction- and parameter-level diversity (an adversary must break both), not hardness-family-disjoint insurance; that comes only from the hash-based Magnetar kind (kind $0x03$, SHA-3 collision). Corona is the Module-LWE parameter-diversity leg of the LP-217 PQ-strict posture, calibrated at 5 000 000 gas. Performance numbers are tagged **measured** or **projected**. Corona's threshold library has landed (`corona v0.10.x`; the dedicated Corona precompile at `0x012206` is wired); the *P3Q-family* kind-$0x02$ dispatch entry at `0x012205` remains reserved and returns `abiFalse` via `ErrKindNotImplemented` until that dispatch is wired, and GPU dispatch is projected, not measured.

2 Corona Kind (kind $= 0x02$)

2.1 Slot and ABI

`0x012205` with leading `kind=0x02`. Same physical precompile address as Pulsar (kind $0x01$) and Magnetar (kind $0x03$); dispatch resolves to the Corona `P3QVerifier` implementation.

```
// EVM precompile at slot 0x012205, kind = 0x02
bool ok = P3Q.verify(
    0x02,                       // Corona kind
    coronaSig,                  // serialized Corona threshold sig
    messageHash,                // H(prev_root || new_root || batch_hash)
    groupPubKey                 // Corona group public key
);
```

2.2 Underlying scheme

Corona [3] is the Lux deployment of a two-round lattice threshold signature scheme building on the Ringtail line [1]. Hardness reduces to Module-LWE over the ring $R_q = \mathbb{Z}_q[X]/(X^{256}+1)$ at the Ringtail parameter set (module matrix $A \in R_q^{8\times 7}$, 48-bit prime q), via the standard two-round distributed signing protocol.

Dealerless key generation. Corona is *natively dealerless*. Key generation runs a Pedersen-style verifiable-secret-sharing DKG over R_q (the `keyera` bootstrap over the shared `luxfi/dkg` VSS: commitment $\mathbf{C} = \mathbf{A}\cdot\mathsf{NTT}(c) + \mathbf{B}\cdot\mathsf{NTT}(r)$, commitment broadcast, a digest cross-check round, and verification with signed-complaint identifiable abort), noise-flooded so the group key is a true $\mathbf{As}+\mathbf{e}''$; no single party ever holds the group secret. As of `corona v0.10.x` this is the production keygen default (`keyera.Bootstrap`), now consuming the shared `luxfi/mlwe` Module-LWE base. Signing likewise never reconstructs the key: each party emits a *masked* partial $\mathbf{z}_j$ from its own share and the combiner sums them ($\mathbf{z} = \sum_j \mathbf{z}_j$), so the master secret $\mathbf{s} = \sum_j \lambda_j \mathbf{s}_j$ appears only as the coefficient of the public challenge in the masked aggregate, never formed.

Pulsar is now also dealerless. Its production FIPS-204 keygen uses Mithril short replicated secret sharing [2] — the published escape from the additive-S_η wall that earlier forced a trusted dealer — and its signing is the no-reconstruct hyperball signer (LP-073 [4]). So in the Quasar dual-PQ AND-mode certificate (LP-020) *both* lattice legs are dealerless: Corona composes alongside Pulsar as **defence-in-depth**, not as the sole carrier of the permissionless-public guarantee. Corona's dealerlessness is unconditional; Pulsar's carries a labeled FIPS-204 KeyGen distribution-equivalence residual (verifier-compatibility is proven), which is one more reason a sovereign validator set runs both legs.

Corona's signature is *not* byte-equal to any NIST standard. Both Corona and Pulsar (FIPS-204 ML-DSA-65) are Module-LWE schemes; they differ in *parameter regime and codebase*, not in hardness family. Corona uses the Ringtail-derived parameter set (∼33 KB signatures, NIST Cat-1) with Module-SIS polynomial commitments; Pulsar uses the ML-DSA-65 tuning (∼3.3 KB, Cat-3, byte-equal to FIPS-204). Pairing them gives parameter- and implementation-level diversity — a break specific to one parameter set or codebase need not hit the other — but *not* family-disjoint hardness: a structural break of Module-LWE would threaten both. Cross-family independence on the signature lane comes only from the hash-based Magnetar kind (kind $0x03$, SLH-DSA).

2.3 Wire format

Offset	Size	Field
0	1	`kind = 0x02`
1	1	`mode = 0x01` (reserved Corona parameter set)
2	4	`sigLen` (BE32)
6	`sigLen`	`coronaSig` (~33 052 B)
6+*sigLen*	4	`pkLen` (BE32)
10+*sigLen*	`pkLen`	`groupPubKey`
10+*sigLen*+*pkLen*	32	`messageHash`

Table 1: Corona-kind wire format. Sig size is the dominant cost (∼ 33 KB); the rest mirrors the Pulsar layout to keep parsing uniform across kinds.

The mode byte is reserved at `0x01`; future Corona parameter set additions (a higher-security Module-LWE variant) get new mode bytes without changing the wire-format skeleton.

2.4 Verifier core

The Corona verifier core IS `luxfi/corona/sign.Verify`, and it is wired today behind the *dedicated* Corona precompile at slot `0x012206` (`precompile/corona/contract.go`, which imports `luxfi/corona/sign` and `luxfi/corona/threshold`). The core is bespoke — not FIPS-204 byte-equal — because Corona is a separate Module-LWE construction (Ringtail/Raccoon lineage), not a NIST standard.

Status: the Corona threshold library has landed (`corona v0.10.x`) and the dedicated Corona precompile at `0x012206` is wired. The *P3Q-family* kind-$0x02$ dispatch at the unified slot `0x012205` is what remains pending: the reference `p3q.Run` still routes `kind=0x02` to `abiFalse` via the `ErrKindNotImplemented` path,

```
case KindCorona, KindMagnetar:
    // Reserved kinds -- P3Q dispatch not yet wired. Return abiFalse
    // ...
    return abiFalse(), remaining, nil
```

so the verifier is callable today at `0x012206` but not yet via the unified P3Q slot. Solidity callers MAY emit `P3Q.verify(0x02, ...)` from network genesis; the call charges the per-kind gas (see below) and returns `false` until the kind-0x02 dispatch is wired. This is the LP-218 "no revert on cryptographic failure" contract: `false` on the return word is the universal signal for "not verified for any reason," and reserved-but-not-wired joins "malformed input" and "FIPS reject" under that umbrella.

2.5 Gas cost

5 000 000 gas – ∼100× the Pulsar kind's 50 000. Returned by the Corona `P3QVerifier.GasCost()` entry. Why the disparity?

1. **CPU verify cost is heavier.** The Corona verifier requires NTT inverse over the Module-LWE ring + polynomial multiplication mod q over the larger Ringtail parameter set (~33 KB sig vs Pulsar's 3.3 KB).
2. **Calibration uses the CPU verify cost as the baseline.** LP-218's calibration discipline assumes a CPU-class verifier – validators are not required to have Blackwell GPUs. Under that assumption, Corona's effective per-validator verify cost is the parity audit's measured ∼1.6 ms CPU, which is approximately 20× the Pulsar CPU verify time (~80 µs). The gas charge is rounded up to 100× Pulsar's for conservative accounting: 5 M gas.
3. **Block-budget headroom.** At 12 M block gas and 5 M per Corona verify, the limit is ∼2 Corona verifies per block – this is the bottleneck for very high- density PQ-strict rollup throughput. The cost cliff is intentional; the LP-217 mode ladder uses it to price cross-family insurance correctly.

Operators with all-Blackwell validator sets may petition for a **Blackwell-discounted Corona gas charge** in a future amendment; LP-220 does not codify it. The conservative CPU-baseline gas charge applies until validator-hardware uniformity is established. This is the same calibration discipline LP-218 uses for the Pulsar kind.

2.6 Performance

Metric	M1 Max	Spark	Source
Per-validator Corona sign at N=64	∼30 ms	∼30 ms	matrix bench (measured) [7]
CPU verify (single sig)	∼1.6 ms	∼1.6 ms	parity audit (measured)
Blackwell sm_120 verify	—	∼15 ms	projected (LP-203 kernel) [5]
Blackwell sm_120 batched (N=64)	—	∼50 ms aggregate	projected

Table 2: Corona kind performance. Sign and CPU-verify times are measured per the parity audit; Blackwell verify timings remain projections until the GPU dispatch boundary closes (see LP-223).

2.6.1 Scaling exponent

The parity audit measured Corona-leg signing cost growing super-linearly in N. The measured exponent across $N \in \{16, 32, 64\}$ is approximately $T(N) \approx c \cdot N^{1.5\text{–}1.7}$, depending on the host platform. This is not an $O(N)$ scheme and should not be described as such in downstream documents; the Module-LWE distributed signing protocol's full-rank check and Gaussian-hash PRF carry costs that grow faster than linearly in the participant count. The measured exponent is the operative number for sequencer capacity planning, not the naive-amortization figure.

2.7 Use case

A rollup that wants **intra-lattice parameter and implementation diversity** configures its sequencer to sign each batch with both a Pulsar threshold sig AND a Corona threshold sig —

two Module-LWE schemes at distinct parameter sets and codebases. The rollup contract on Z-Chain calls `P3Q.verify(0x01, ...)` and `P3Q.verify(0x02, ...)` at the same precompile address `0x012205` in sequence inside its `validateBatch` function; both must return `true` for the batch to be accepted.

A break specific to Pulsar's parameter set or codebase need not compromise Corona's, so the rollup retains a valid PQ leg under an implementation- or parameter-localized failure. This is *not* family-disjoint hardness: a structural break of Module-LWE itself would threaten both legs. For cross-family insurance the rollup must add the hash-based Magnetar kind (`P3Q.verify(0x03, ...)`). The cost is $\sim$5 050 000 gas per batch (Pulsar 50 000 + Corona 5 000 000). Acceptable for value-bearing rollups (custody, treasury, regulated securities); prohibitive for high-volume rollups (gaming, social).

This is the LP-217 `PQ-strict` posture made operationally callable.

3 Application: DEX Rollup Settlement to Z-Chain

The DEX matching path (Lux Lightspeed DEX [8]) fills orders at hundreds of millions per second per GPU on the D-Chain, far above any single chain's finality rate. To anchor that throughput trustlessly, per-market fill batches roll up to the Z-Chain and are verified there through the P3Q family at slot `0x012205`. This section places Corona (kind $0x02$) in that settlement pipeline; the measured verify rates belong to the sibling kinds, and Corona is the optional Module-LWE parameter-diversity leg.

3.1 Two verification rungs

DEX rollup settlement uses a two-rung ladder, both callable at the same P3Q slot:

1. **Attestation rung (venue-trust).** The sequencer signs each per-market batch with a P3Q kind $0x01$ (Pulsar / FIPS-204 ML-DSA-65) threshold signature. The Z-Chain contract calls `P3Q.verify(0x01, ...)` to confirm the batch was produced by the venue's threshold quorum. This rung trusts the venue's signer set but is cheap: the ML-DSA-65 verify path is measured at $\sim$4714 verifies per second.
2. **Validity rung (trustless).** For trust-minimized settlement the sequencer additionally produces a STARK/FRI validity proof of correct matching, verified by the `starkfri` verifier (LP-221 [6]) — a cSHAKE256 / Goldilocks PQ Plonky3-fork verifier. This rung proves the fills are correct without trusting the signer set. The current verifier is measured at $\sim$986 proofs per second against a **toy AIR**; this figure is the operative number and is explicitly not a production-circuit measurement.

The attestation rung is fast and venue-trusting; the validity rung is slower and trustless. A venue chooses the rung (or runs both) per its trust requirements.

3.2 Where Corona fits

Corona (kind $0x02$) is the *Module-LWE cross-family insurance leg* for the attestation rung, not a separate settlement mechanism. A value-bearing DEX rollup that wants independence from the Module-LWE assumption underlying ML-DSA-65 signs each batch with *both* a Pulsar (kind $0x01$) and a Corona (kind $0x02$) threshold signature and requires both `P3Q.verify` calls to return `true`, exactly as in Section 2's use case. A Module-LWE break that compromises the Pulsar attestation does not compromise the Corona attestation; the rollup's settlement remains cryptographically anchored.

The cost discipline of Section 2 applies unchanged: Corona's 5 000 000 gas makes the second leg appropriate for value-bearing markets (custody, treasury, regulated-securities venues) and prohibitive for high-volume markets, which settle on the kind $0x01$ attestation rung alone (optionally backed by the starkfri validity rung). As noted throughout this paper, Corona's GPU dispatch is projected and its *P3Q* kind-$0x02$ dispatch is not yet wired (the threshold library has landed and the dedicated Corona precompile at `0x012206` is live). A DEX settlement path that wants a second Module-LWE scheme (distinct parameter set and codebase, not a distinct hardness family) uses the measured kind $0x01$ rung today, with Corona reachable at `0x012206` and reserved at the P3Q kind-$0x02$ entry until that dispatch is wired.

3.3 Bundled settlement: $K \times V \times$ markets

Per-fill, per-venue verification does not scale: at the attestation-rung's ∼4714 verifies per second, a chain that verified one signature per fill would cap settlement far below the matcher's fill rate. The settlement model therefore bundles. A single Z-Chain settlement object aggregates K signer attestations across V venues across many markets into one verification event:

- **K signers.** The threshold quorum per venue contributes K partial signatures aggregated into one threshold signature that verifies in a single `P3Q.verify` call.
- **V venues.** Independent venues' batches are aggregated into one settlement object rather than verified one venue at a time.
- **Many markets.** Each venue's per-market arenas (Lux Lightspeed DEX) contribute their batch roots to the bundle, so a single verification covers many books.

Because one verification amortizes over $K \times V \times$ markets fills, the effective settled-fill rate at fleet scale is **projected** to exceed 10^9 fills per second of finality — this is a fleet-aggregation projection, not a single-verifier measurement. The per-verifier measured rates (∼4714/s attestation, ∼986/s validity on a toy AIR) are the operative numbers; the $>10^9$ figure is what those rates compose to once the bundle factor $K \times V \times$ markets is applied across a multi-node Z-Chain settlement fleet. No single P3Q verifier reaches that rate.

References

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[2] Andrea Celi, Rafaël del Pino, Thomas Espitau, Guilhem Niot, and Thomas Prest. Threshold ML-DSA from short replicated secret sharing (Mithril). In *USENIX Security*, 2026. https://eprint.iacr.org/2026/013. Short replicated secret sharing: the published escape from the naive-additive-S_η wall, making dealerless threshold ML-DSA keygen reachable under the stock FIPS-204 verifier.

[3] Lux Industries. Corona: Two-round module-lwe threshold signatures. https://github.com/luxfi/corona, 2026. Lux's own threshold scheme: Module-LWE (Ringtail/Raccoon lineage), a rank-8×7 module over $R_q = \mathbb{Z}_q[X]/(X^{256}+1)$, $q =$ `0x1000000004A01` — *not* Ring-LWE (rank 1). Two-round, natively dealerless (Pedersen-VSS, noise-flooded), no-reconstruct signing; consumes the shared `luxfi/mlwe` base. Builds on the Ringtail line ([1]); cited via this Lux artifact, never via the academic paper. Companion hardness is Module-SIS.

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